SPLICE PLAYGROUND

Visualize Splicing Into Structure

DNA editing, transcript consequences, and protein structure prediction in one continuous flow.

Start

2026S – Graduation Project2

Splice Playground

Make your own mutant

AI01분반
4조: 강민준, 김현우, 남윤정, 이정균, 최진범
Department of AI-Software

Outline

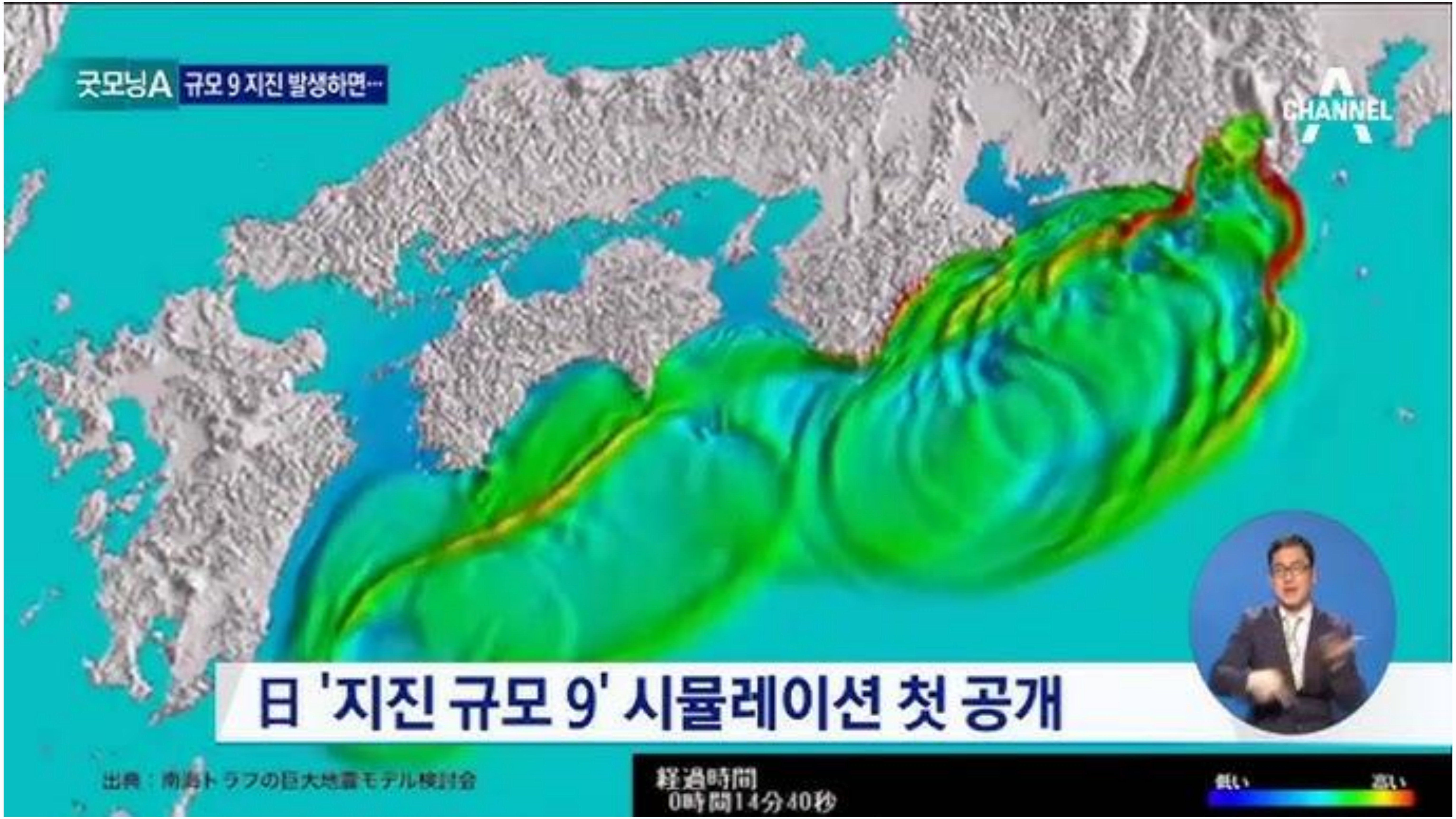
1. Concept Review
2. Splice AI training result
3. Service Architecture
4. DB Construction
5. Progress & Demo

1. Concept Review

Concept Review

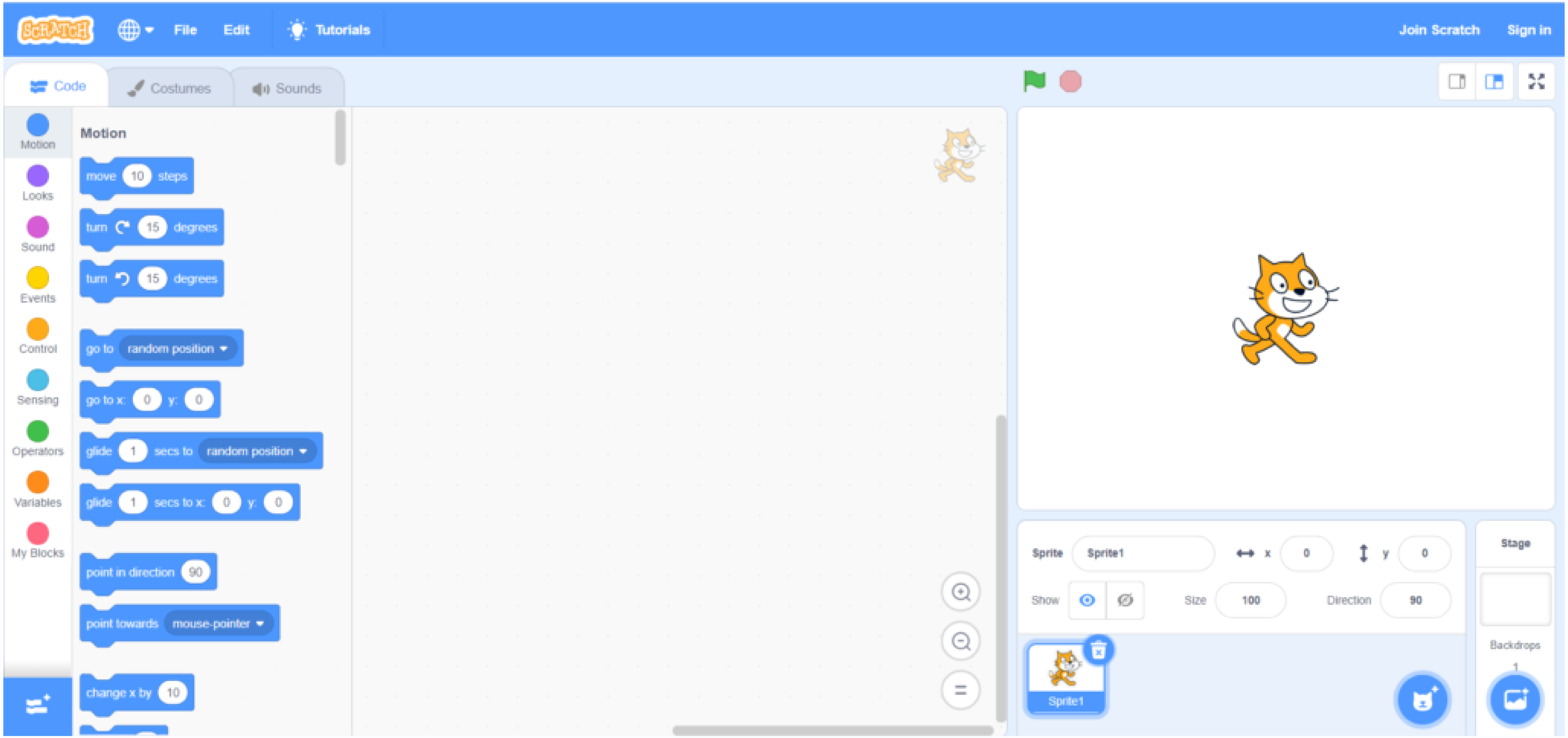
https://www.technologynetworks.com/genomics/articles/genetic-modification-techniques-and-applications-382001

Concept Review



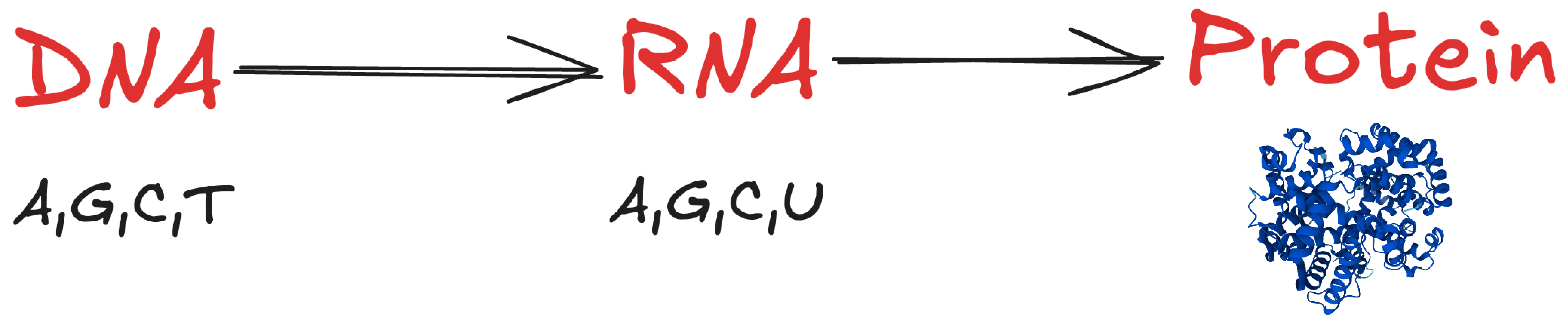
https://www.youtube.com/watch?v=LRCPYVAZF_A

Concept Review

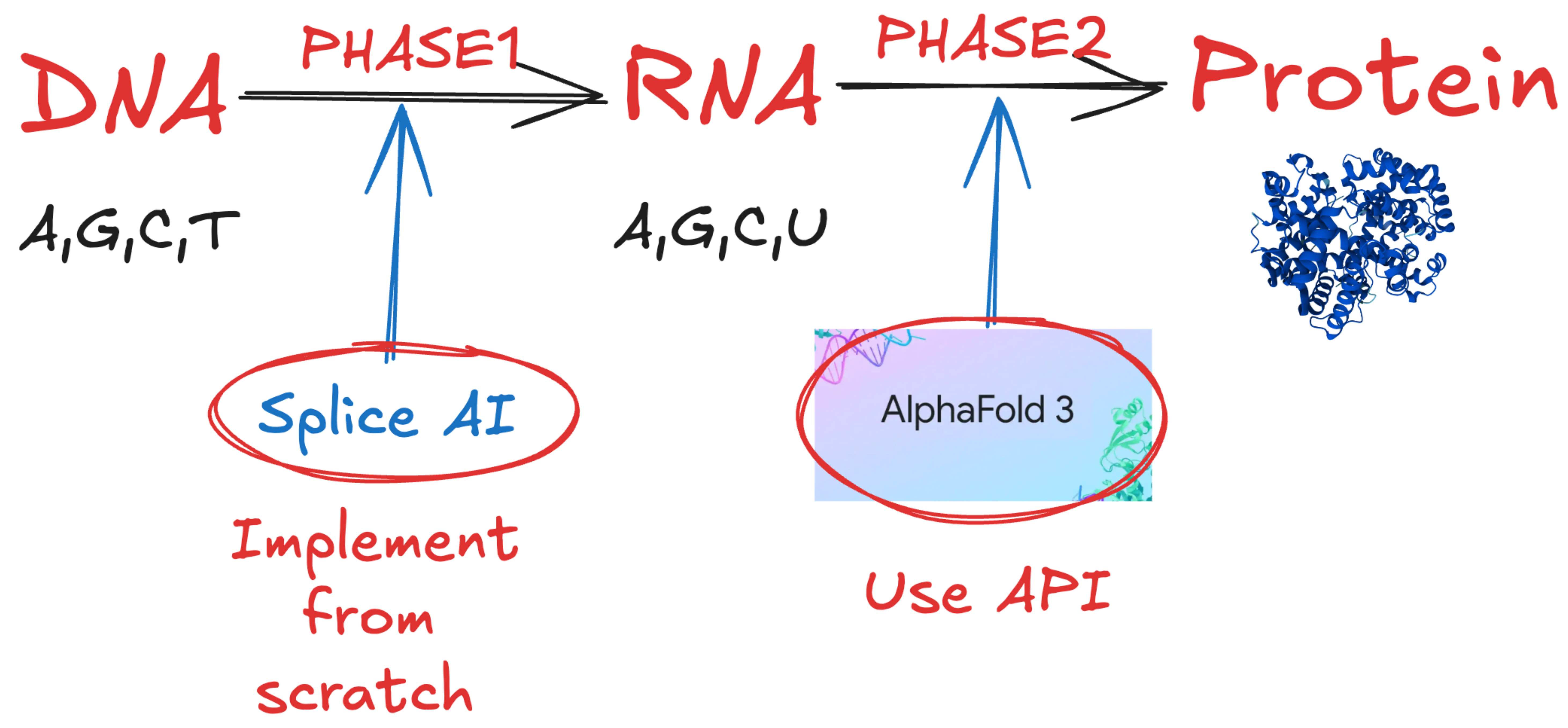


https://www.scienceworld.ca/resource/getting-started-scratch/

Concept Review

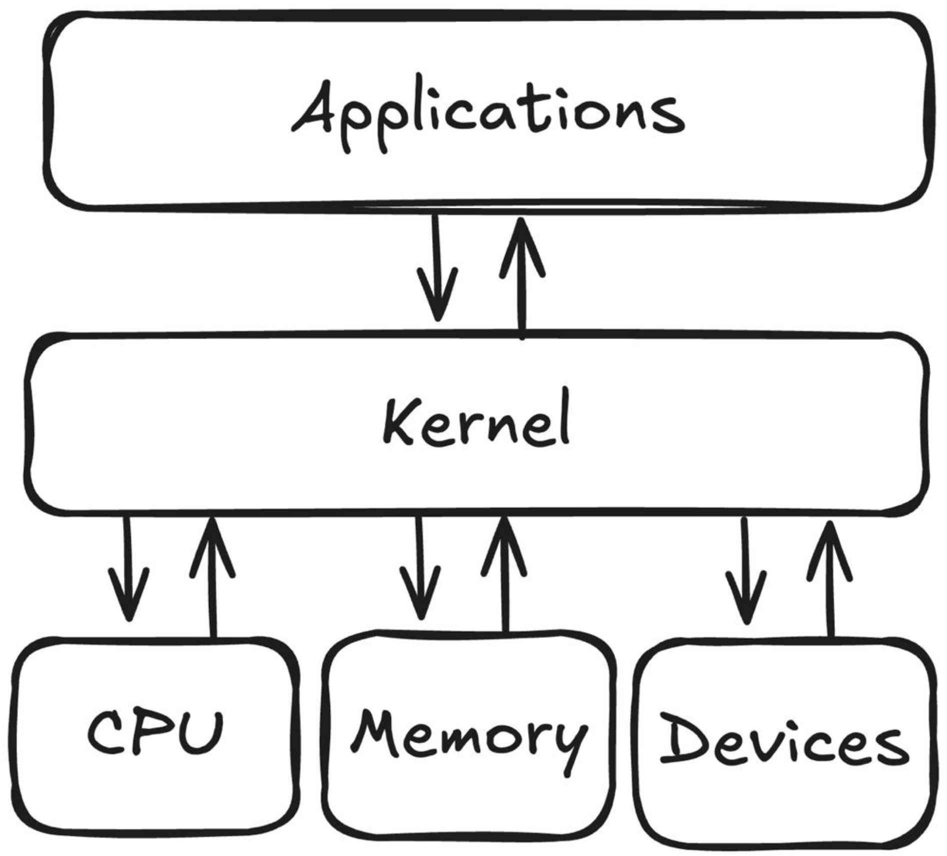


Concept Review



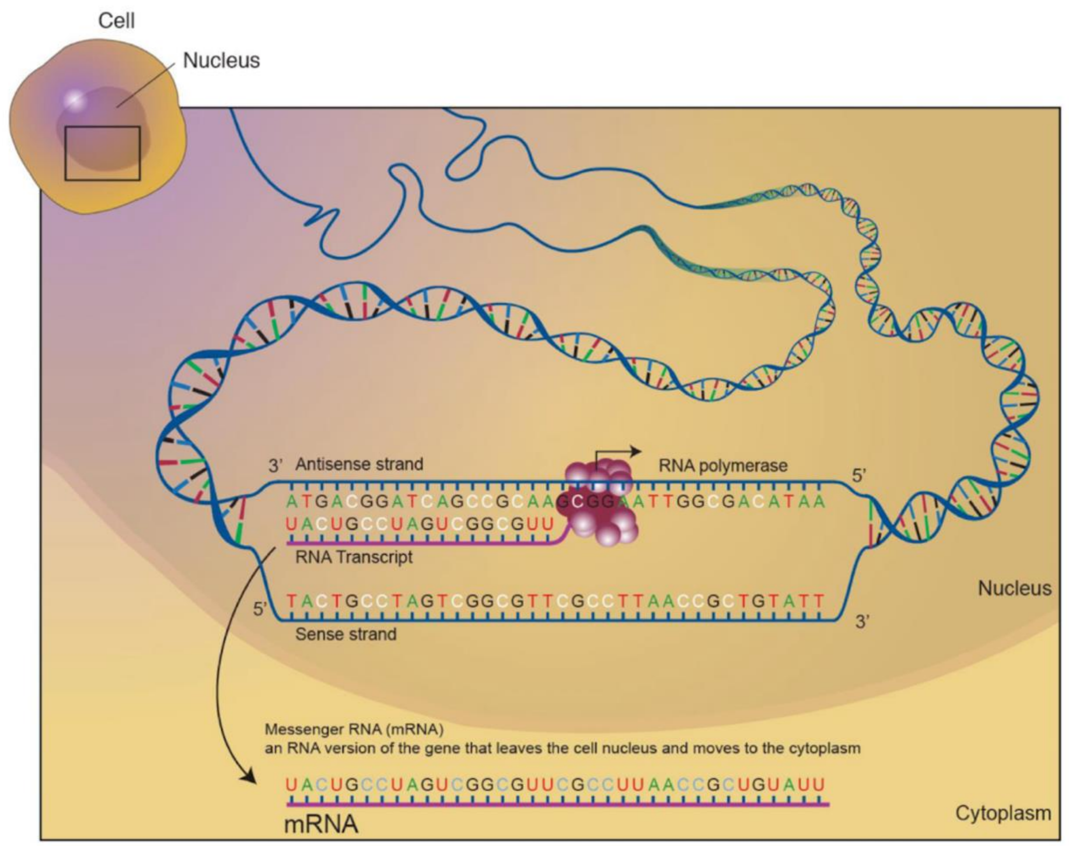
Why AI for Biological Sciences?

OS



Systematic!!

DNA



Unsystematic...

https://www.genome.gov/

What does mean “Unsystematic”?

DNA = kernel + interpreter + PYTHON + library +....

only with A, C, G, T

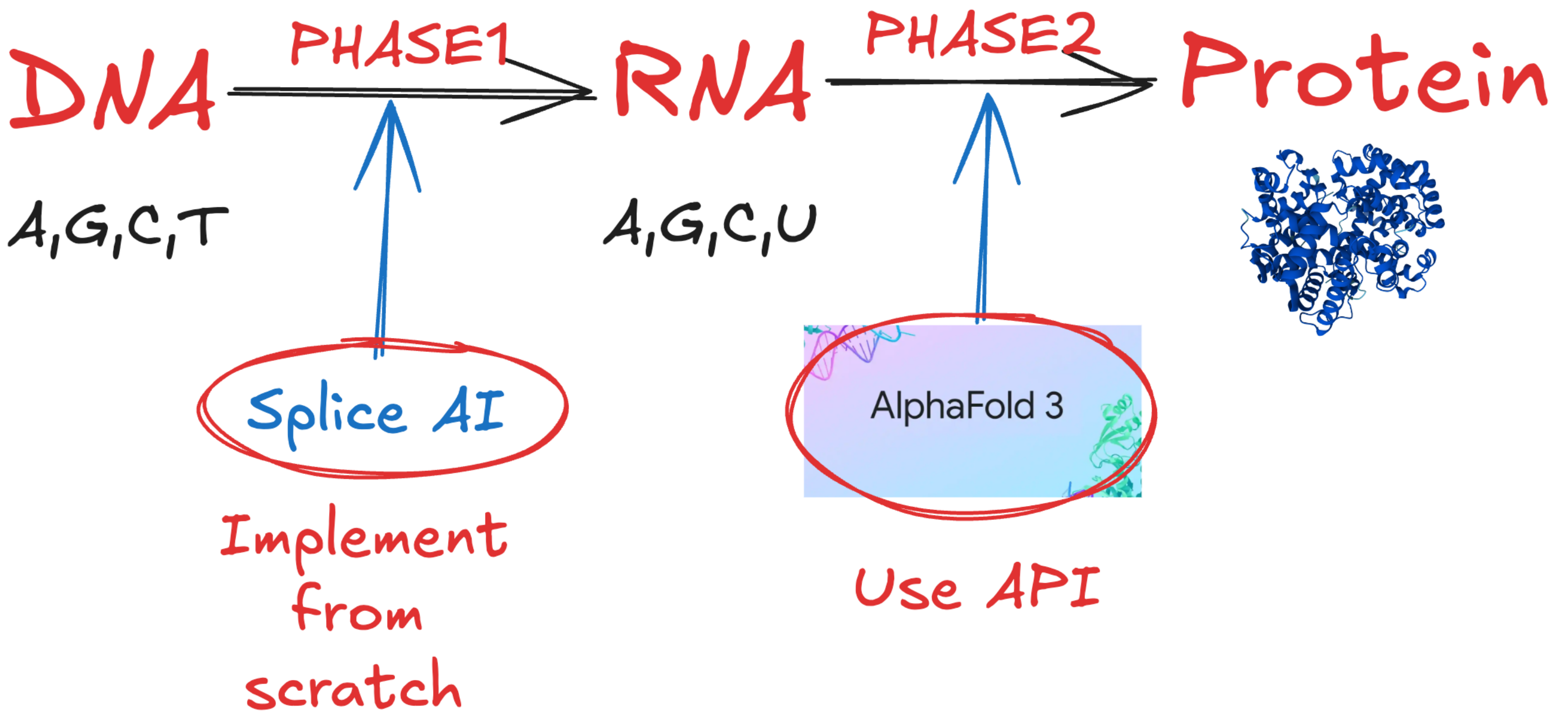
ATGACGGTCAGCCGCAACGGTACTTAAGATTCGCC....

3.2billion characters!!...
32억글자!!!...

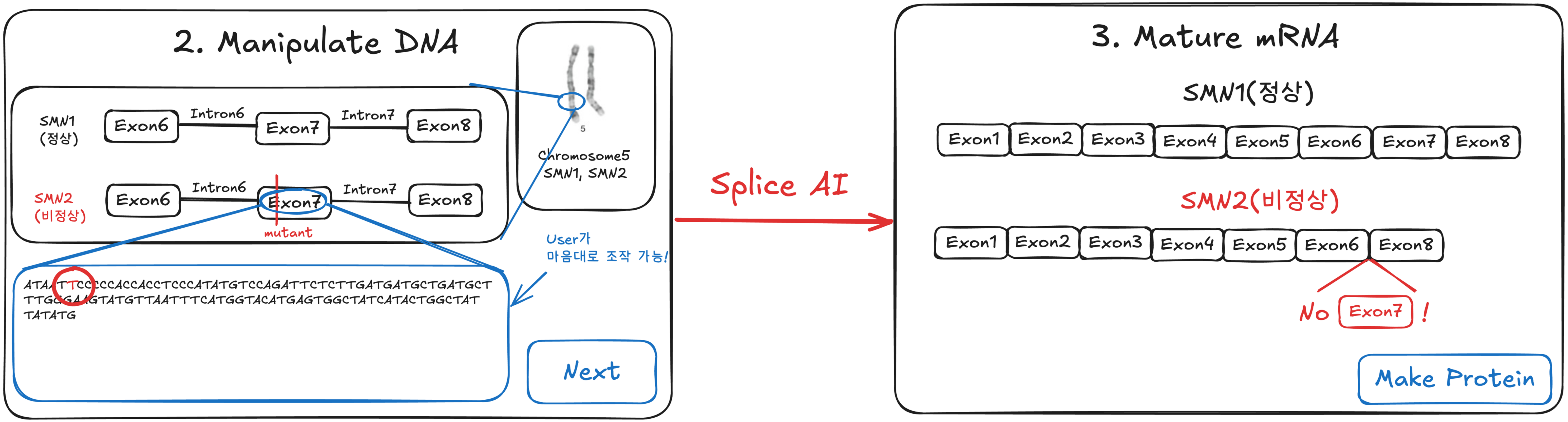
Until today, human can't interpret DNA

2. Splice AI Training Result

Splice AI



DNA
PHASE1
RNA
PHASE2
Protein
A,G,C,T
A,G,C,U
Splice AI
AlphaFold 3
Implement from scratch
Use API



2. Manipulate DNA
SMN1 (정상)
Exon6
Intron6
Exon7
Intron7
Exon8
SMN2 (비정상)
Exon6
Intron6
Exon7
Intron7
Exon8
mutant
5
Chromosome5
SMN1, SMN2
ATAATTCCCCCACCACCTCCCATATGTCCAGATTCTCTTGATGATGCTGATGCT
TTGCGAAGTATGTTAATTTCATGGTACATGAGTGGCTATCATACTGGCTAT
TATATG
User가
마음대로 조작 가능!
Next
Splice AI
3. Mature mRNA
SMN1(정상)
Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon7 Exon8
SMN2(비정상)
Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon8
No Exon7 !
Make Protein

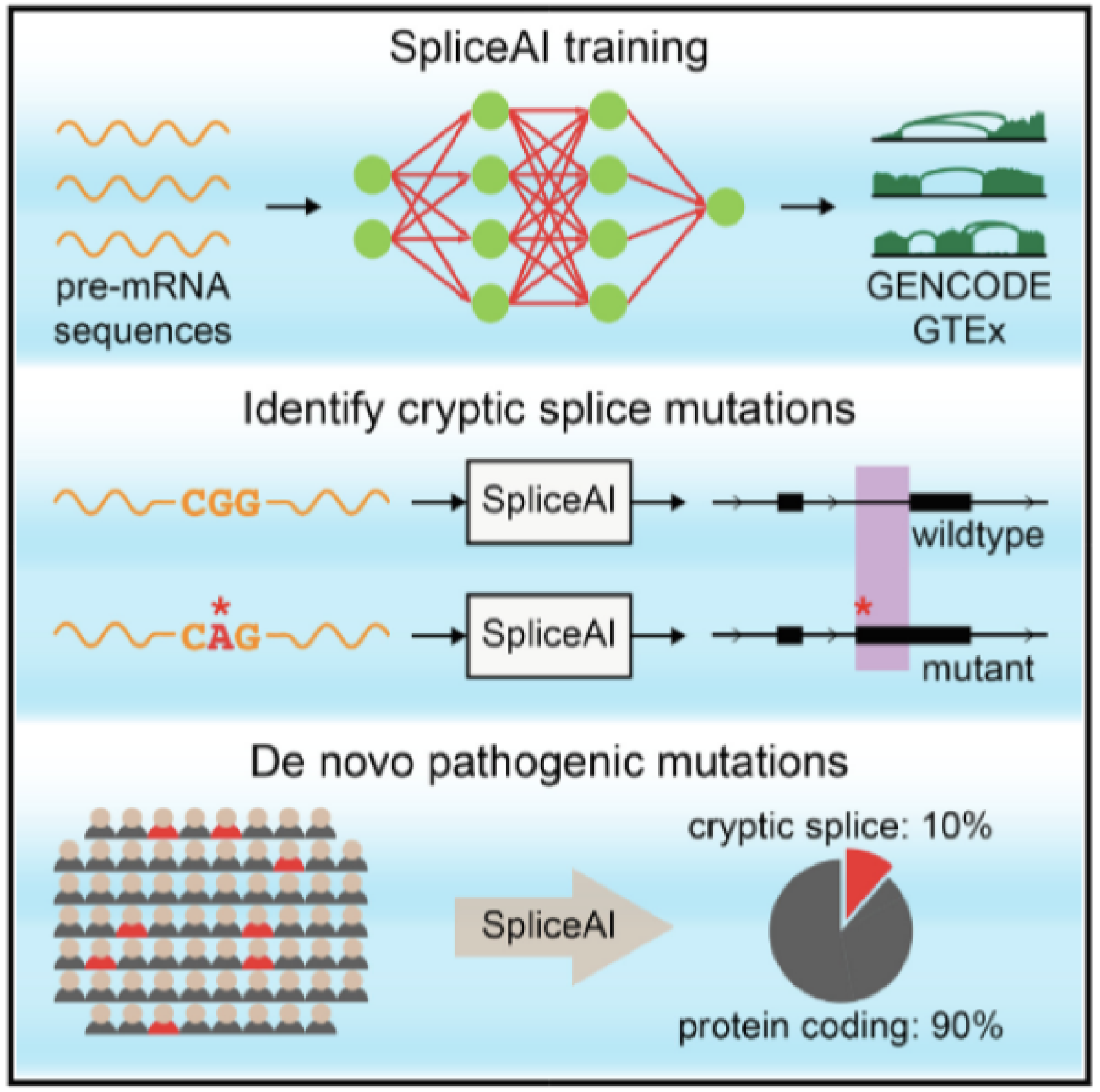
Splice AI

Article

Cell

Predicting Splicing from Primary Sequence with Deep Learning

Graphical Abstract



Authors

Kishore Jaganathan, Sofia Kyriazopoulou Panagiotopoulou, Jeremy F. McRae, ..., Serafim Batzoglou, Stephan J. Sanders, Kyle Kai-How Farh

Correspondence

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In Brief

A deep neural network precisely models mRNA splicing from a genomic sequence and accurately predicts noncoding cryptic splice mutations in patients with rare genetic diseases.

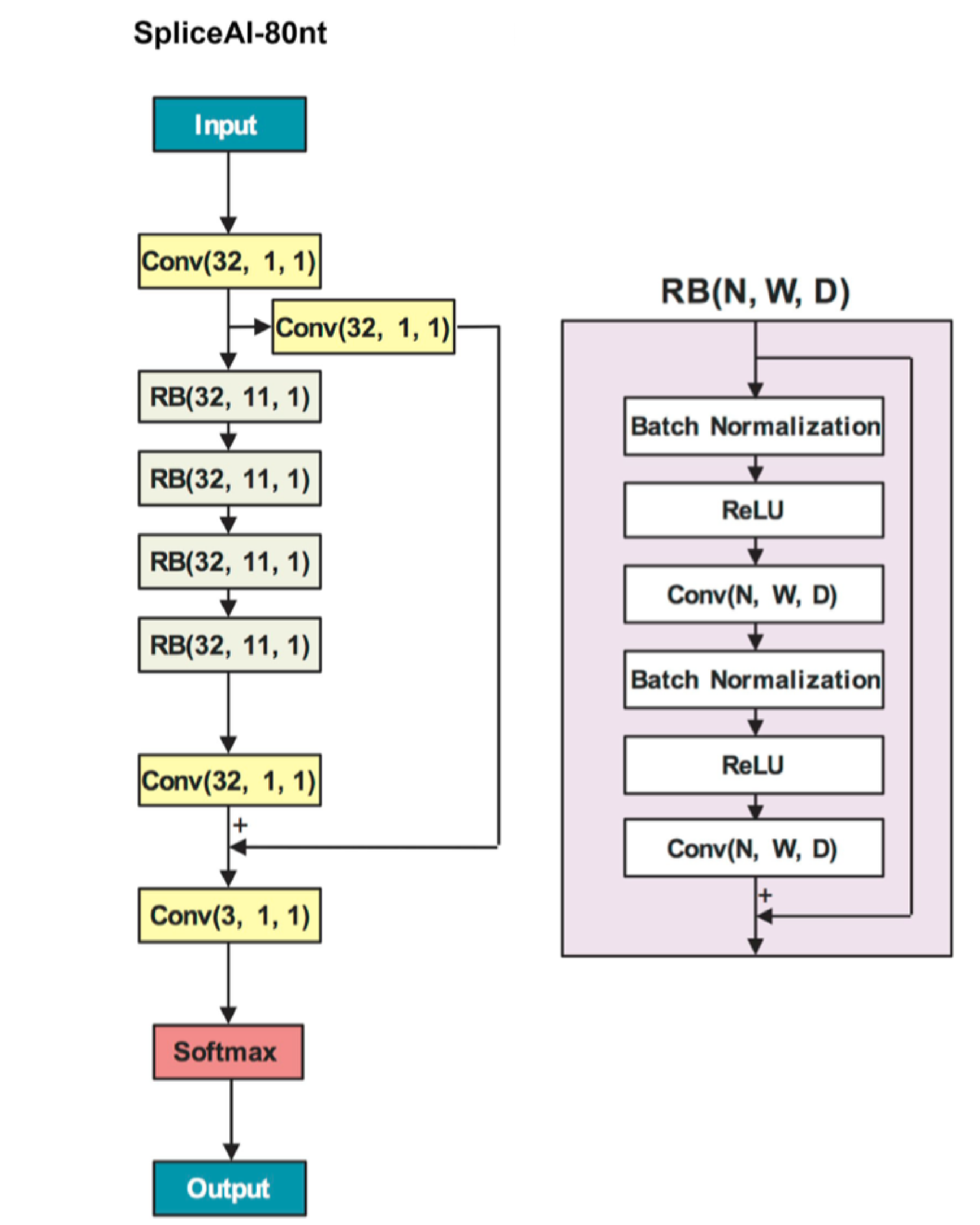
Highlights

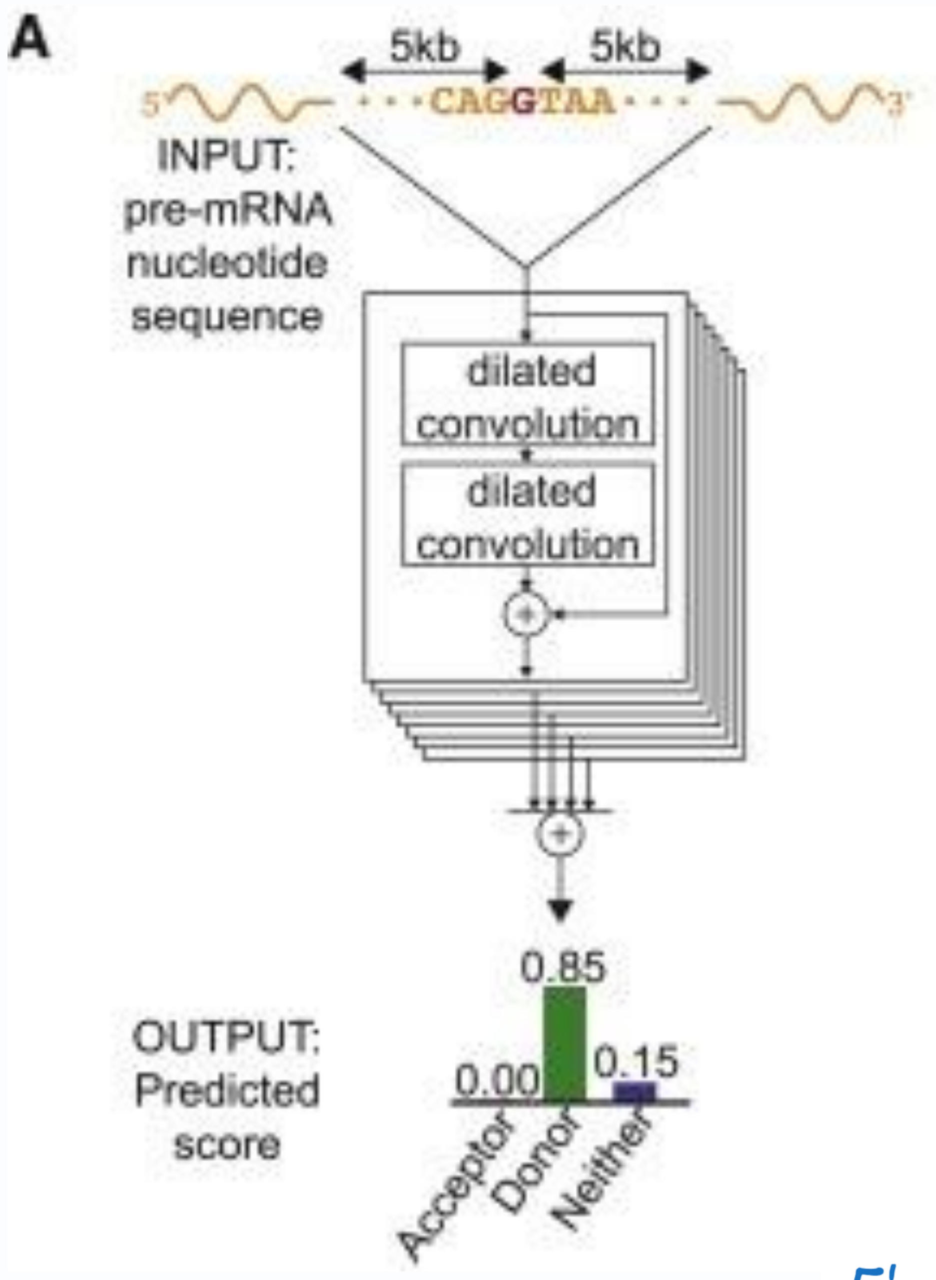
- SpliceAI, a 32-layer deep neural network, predicts splicing from a pre-mRNA sequence
- 75% of predicted cryptic splice variants validate on RNA-seq
- Cryptic splicing may yield ~10% of pathogenic variants in neurodevelopmental disorders
- Cryptic splice variants frequently give rise to alternative splicing

Jaganathan et al., 2019, Cell *176*, 535–548
January 24, 2019
https://doi.org/10.1016/j.cell.2018.12.015

Predicting Splicing from Primary Sequence with Deep Learning. *Cell*. 2019 Jan 24;176(3):535-548.e24.

Splice AI



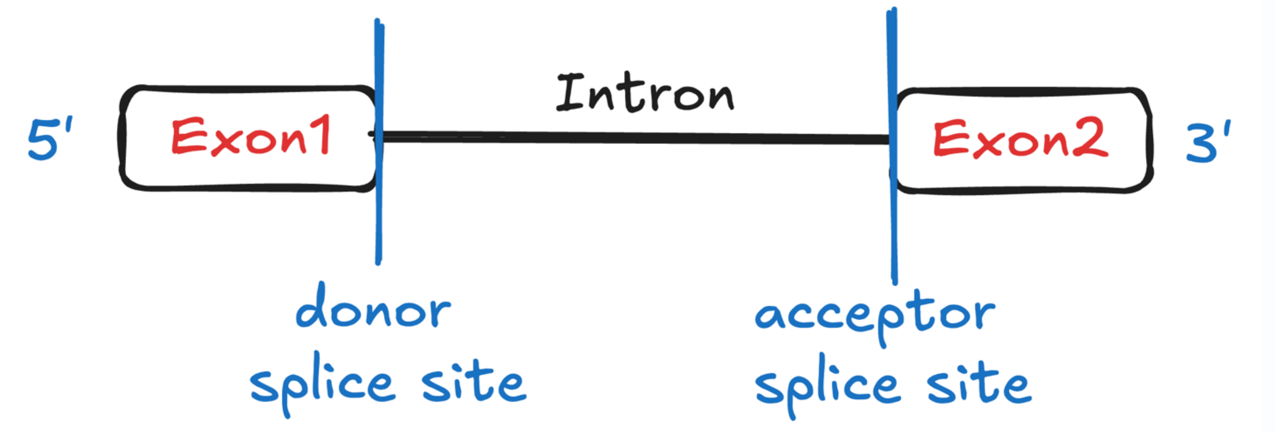


- **Input**: one-hot encoded primary sequence

A	[1,0,0,0]
C	[0,1,0,0]
G	[0,0,1,0]
T	[0,0,0,1]

A	T	G	G	C	T	C	A	T
1	0	0	0	0	0	0	1	0
0	0	0	0	1	0	1	0	0
0	0	1	1	0	0	0	0	0
0	1	0	0	0	1	0	0	1

- **Output**: donor / acceptor / neither (3 classes)



Studied SpliceAI paper at SNU Bio

교수

백대현 BAEK, DAEHYUN

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Email : baek@snu.ac.kr

https://biosci.snu.ac.kr/people/faculty?mode=view&profidx=17

SCIENCE
자 연 과 학 대 학 · 생 명 과 학 부

Took 1st place at SNU

성 적 증 명 서

성 명 : 강민준
생년월일 : 2004. 6. 18.
학 번 : 2025-[redacted]

과 정 : 국내교환
수학대학 : 자연과학대학 생명과학부
수학기간 : 2025. 3. 3. ~ 2025. 6. 14., 2025. 9. 1. ~ 2025. 12. 12.

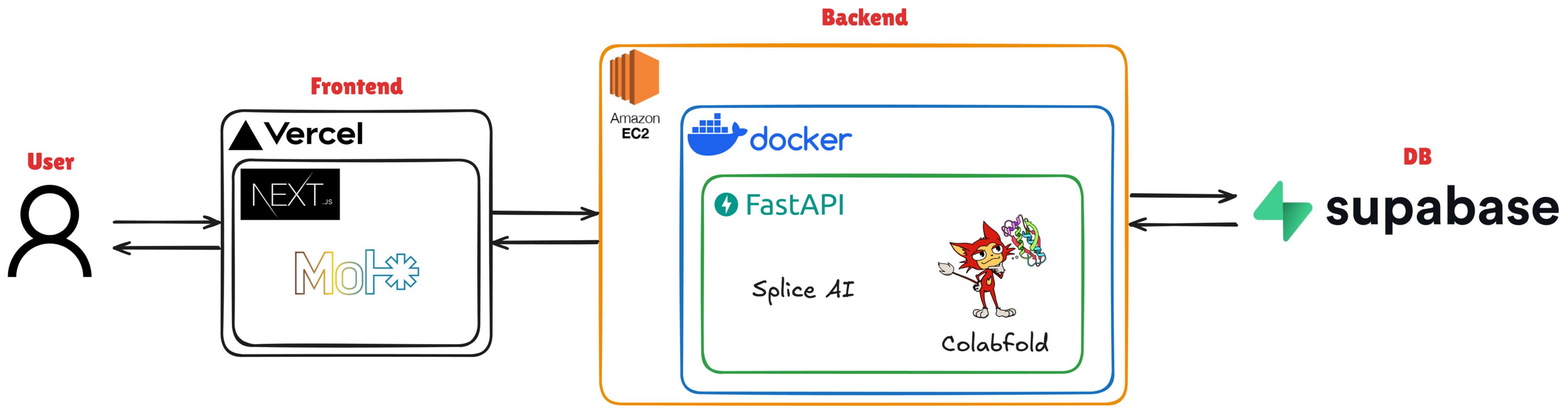
취 득 학 점

학 기	구분	교 과 목 명	학점	성적	학 기	구분	교 과 목 명
2025. 1.	교양	기초생물학 1	1	S			
2025. 1.	교양	생물학실험	1	A+			
2025. 1.	일선	인공지능 개론	3	S			
2025. 1.	대학원	영어논문작성법: 이공계열	1	S			
		학점계 : 6					
		평점평균 : 4.30					
2025. 2.	전선	생명과학을 위한 인공지능	4	A+			
		학점계 : 4					
		평점평균 : 4.30					
		평점평균 : 4.30					
		백점환산점수 : 100.00					

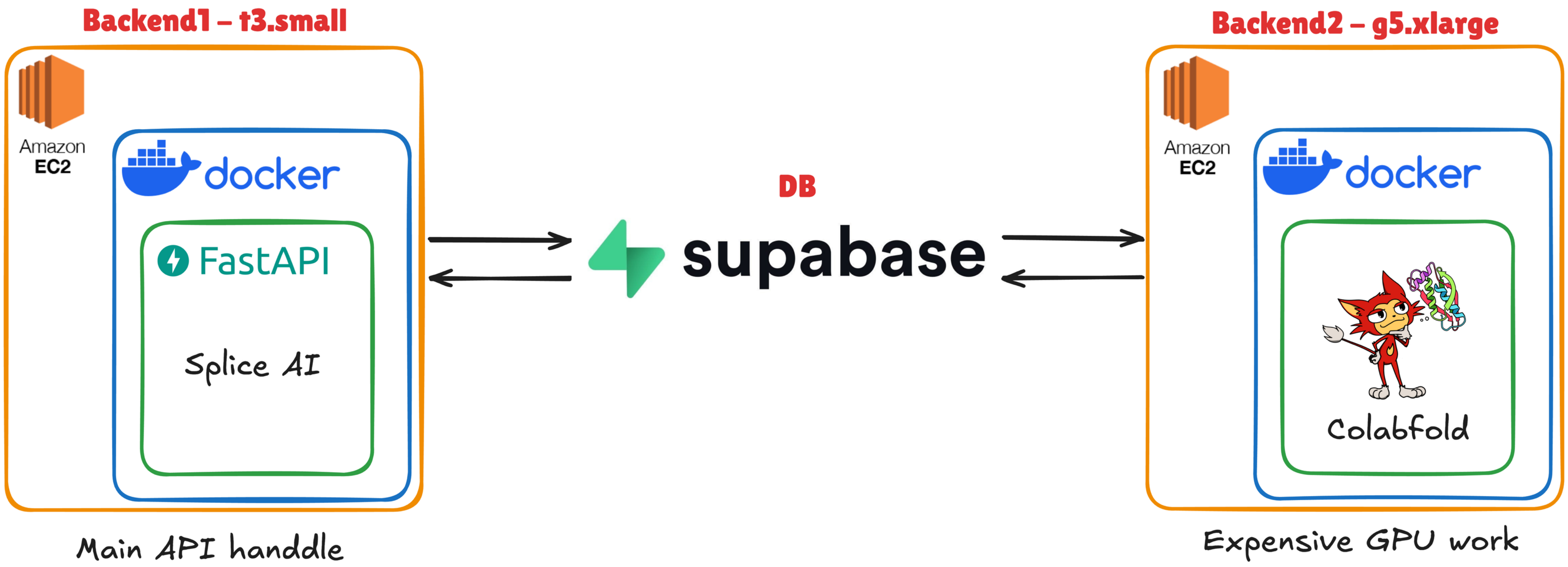
2025. 2. 전선 생명과학을 위한 인공지능 4 A+

3. System Architecture

System Architecture



Backend details – Asynchronous Operation



t3.small
Family: t3 2 vCPU 2 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.0496 USD per Hour
On-Demand Windows base pricing: 0.0392 USD per Hour
On-Demand SUSE base pricing: 0.0518 USD per Hour
On-Demand Linux base pricing: 0.0208 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0243 USD per Hour

g5.xlarge
Family: g5 4 vCPU 16 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 1.013 USD per Hour
On-Demand Windows base pricing: 1.19 USD per Hour
On-Demand RHEL base pricing: 1.0636 USD per Hour
On-Demand SUSE base pricing: 1.0623 USD per Hour
On-Demand Linux base pricing: 1.006 USD per Hour

Backend details – Asynchronous Operation

Backend details – STEP1

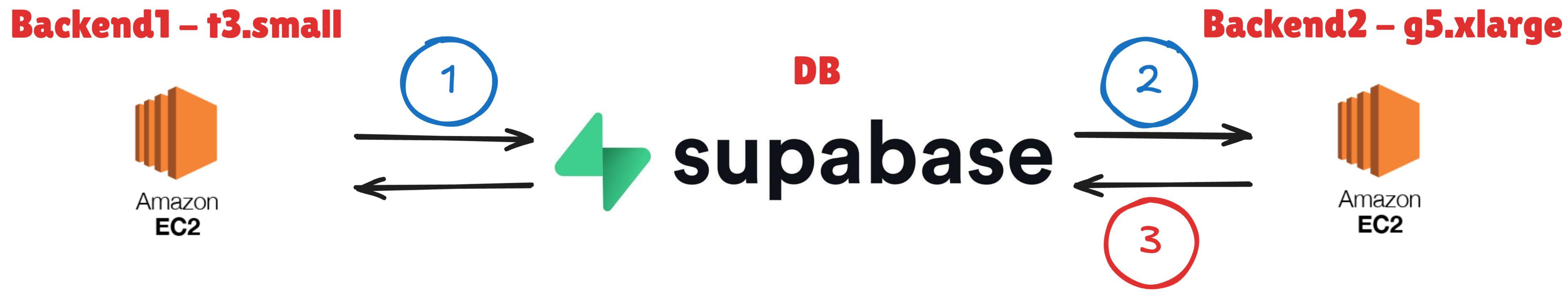
1. GPU연산이 필요 없는 STEP1~3에 대해 API hanndle (Splice AI 포함)
& STEP3에서 나온 결과인 '단백질 서열'을 DB에 업로드

Backend details – STEP2

Backend1 – t3.small **Backend2 – g5.xlarge**

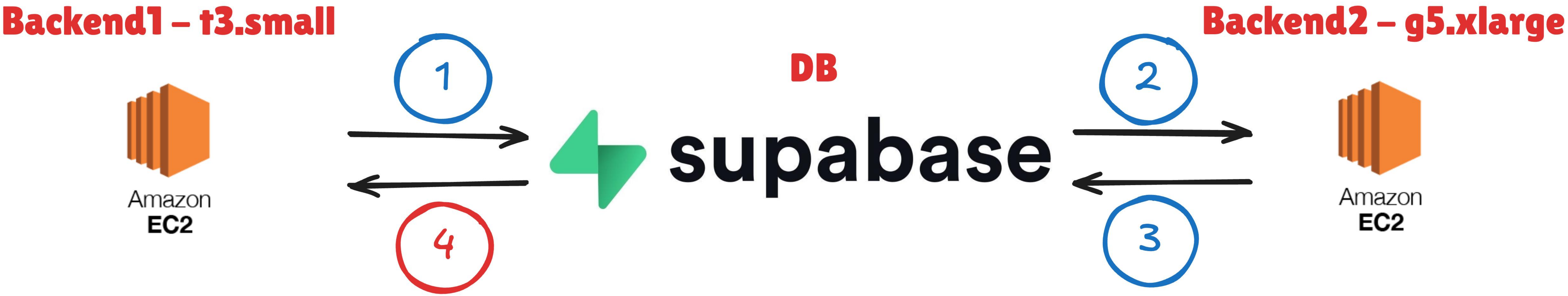
2. DB에 올라온 '단백질 서열'을 읽어서 구조 예측(Colabfold!!)
& 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP3



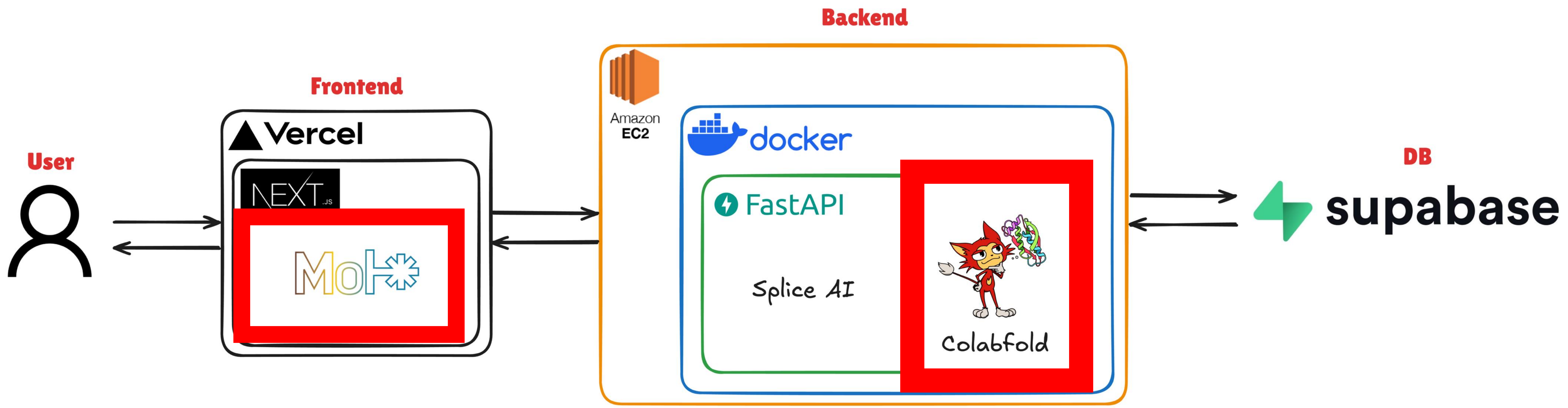
3. & 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP4



4. API hanndling하는 main 백엔드 서버가 단백질 구조 파일을 받아온 후,
프론트엔드에 단백질 구조 파일 반환

System Architecture



AlphaFold -> ColabFold

Brief Communication | Open access | Published: 30 May 2022

ColabFold: making protein folding accessible to all

Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, Lim Heo, Sergey Ovchinnikov & Martin Steinegger

Nature Methods **19**, 679–682 (2022) | Cite this article

VERI LUX TAS MEA

SCIENCE
자 연 과 학 대 학 · 생 명 과 학 부

부교수

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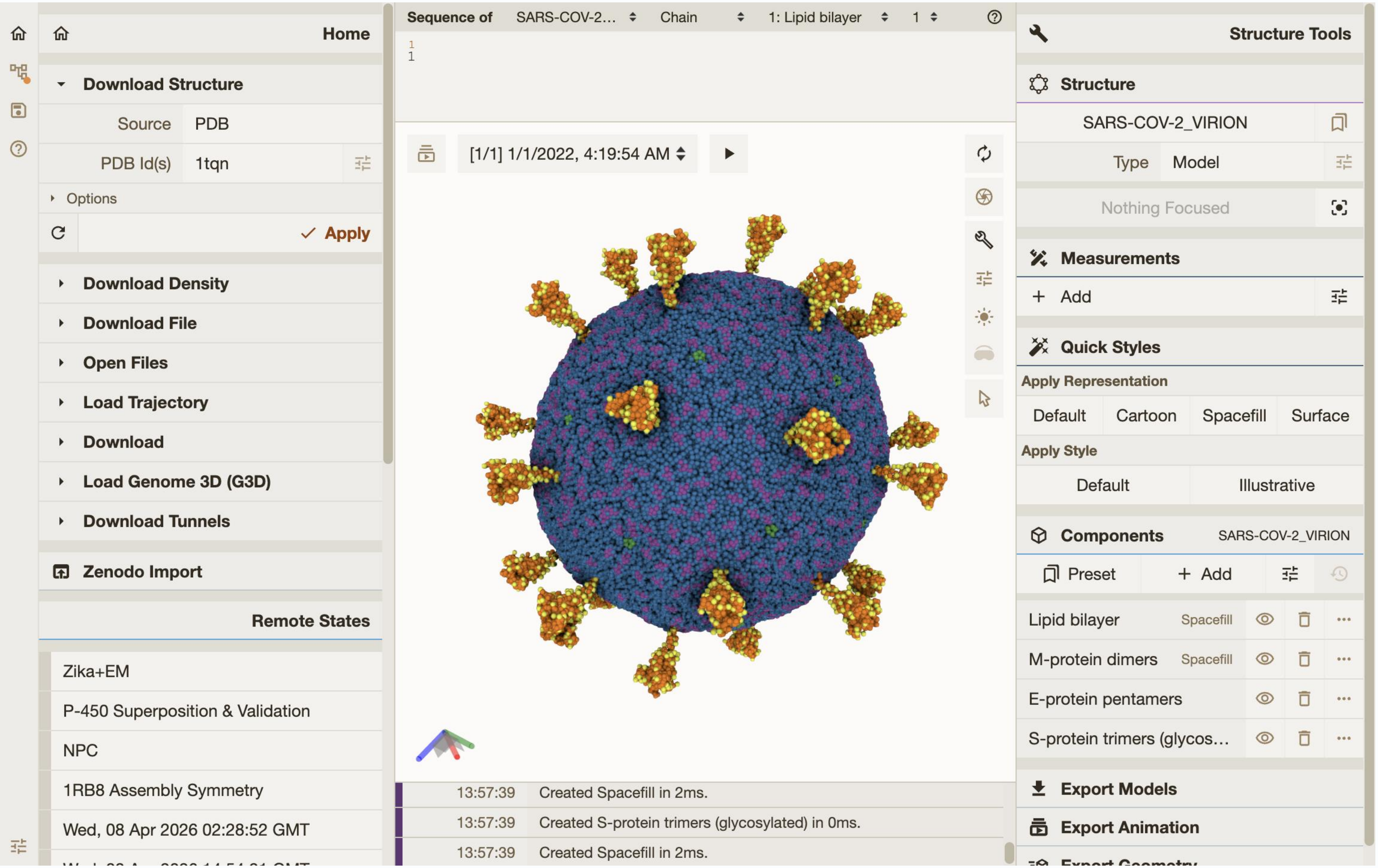
https://www.nature.com/articles/s41592-022-01488-1

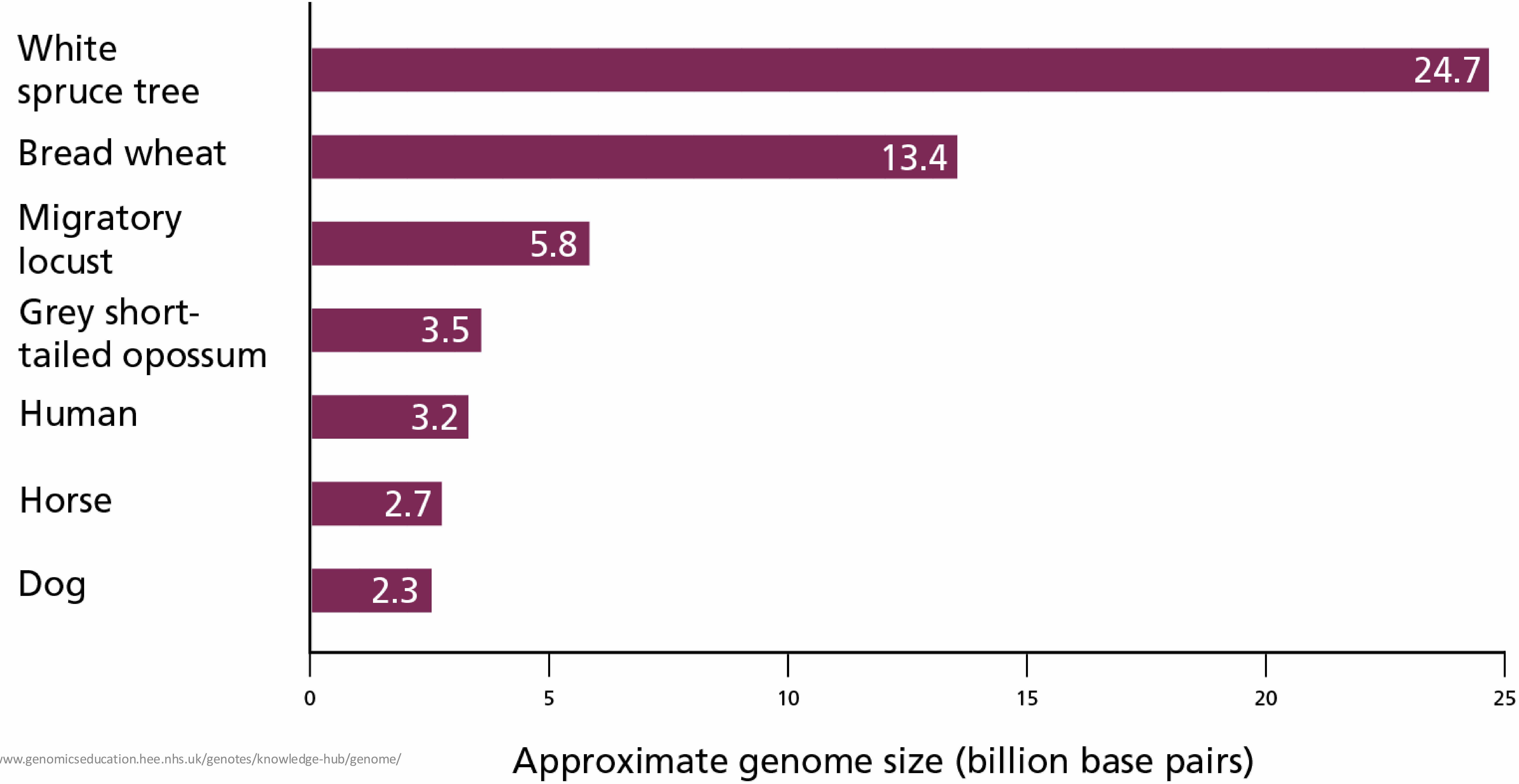
Mol Star



https://molstar.org/viewer/

4.DB Construction

Human Genome Size – 3.2billion!...



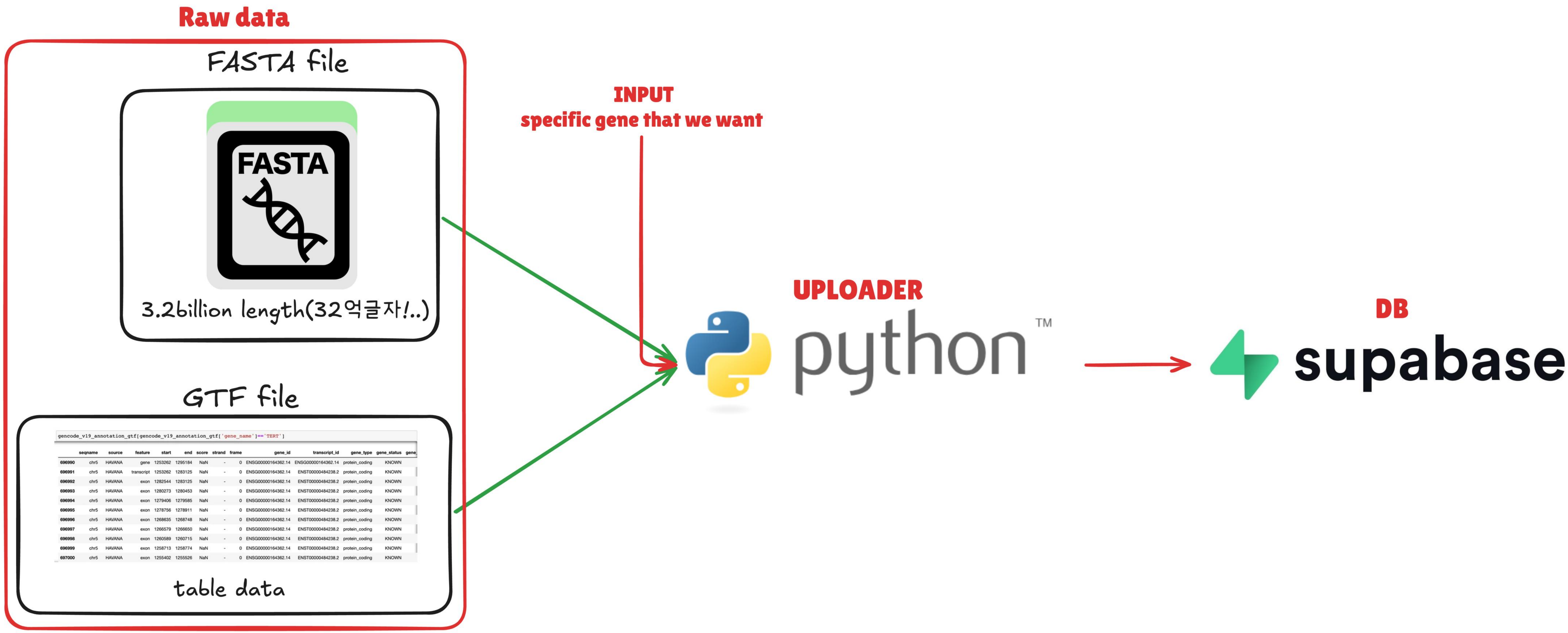
https://www.genomicseducation.hee.nhs.uk/genotes/knowledge-hub/genome/

DB

DB

supabase

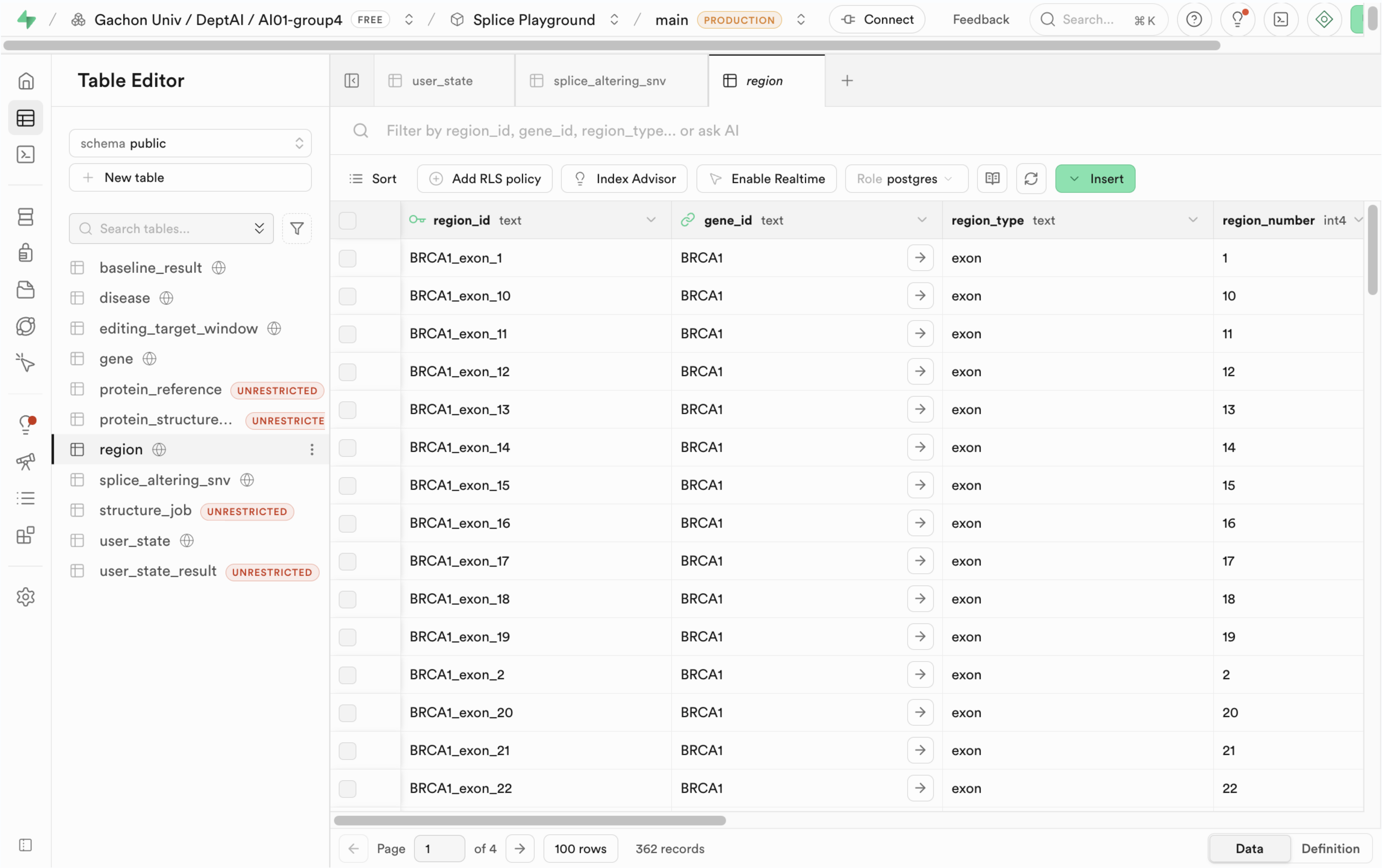
DB upload pipeline



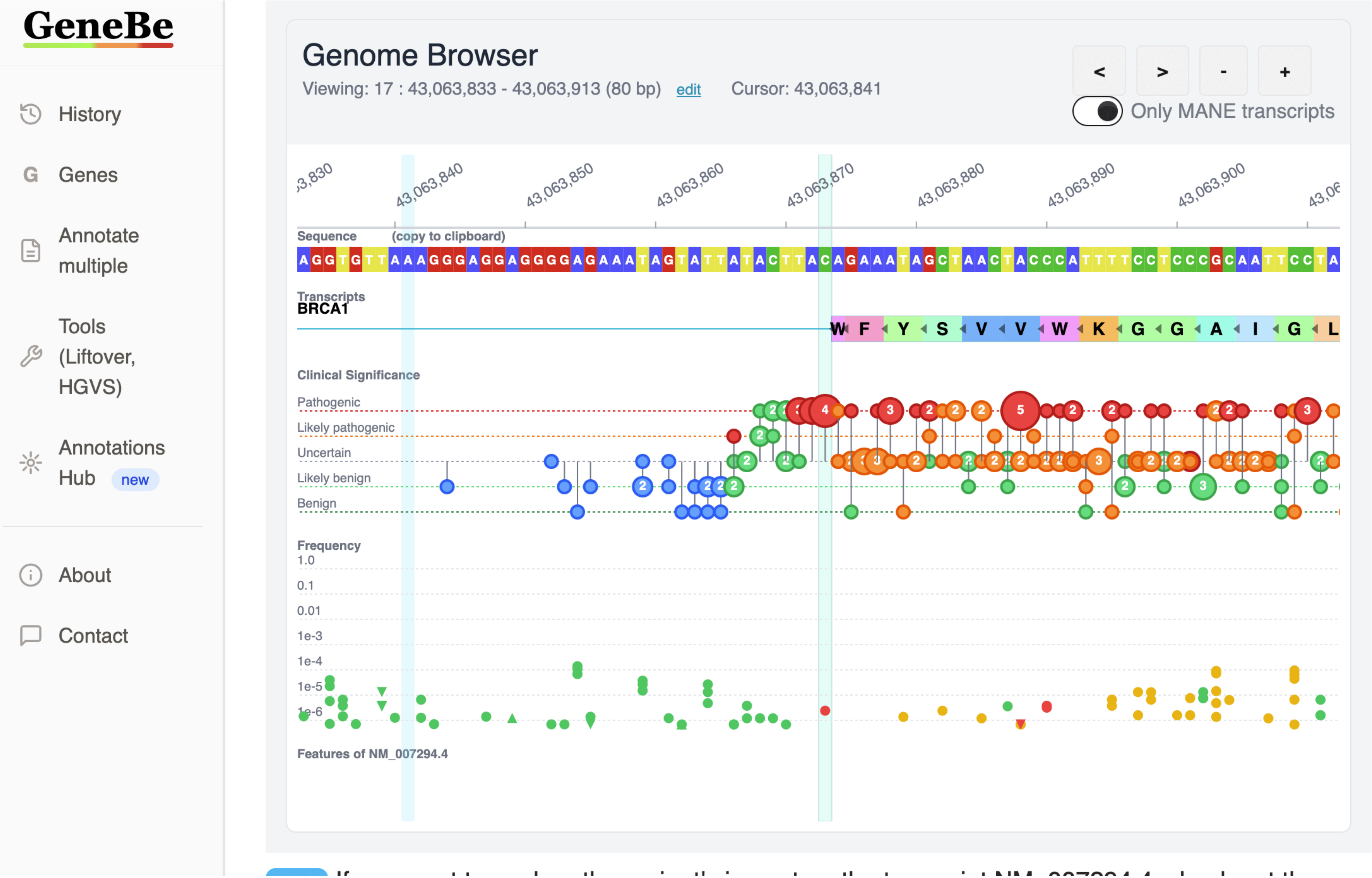

```
gencode_v19_annotation_gtf[gencode_v19_annotation_gtf['gene_name']=='TERT']
```

	seqname	source	feature	start	end	score	strand	frame	gene_id	transcript_id	gene_type	gene_status	gene_
696990	chr5	HAVANA	gene	1253262	1295184	NaN	-	0	ENSG00000164362.14	ENSG00000164362.14	protein_coding	KNOWN	
696991	chr5	HAVANA	transcript	1253262	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696992	chr5	HAVANA	exon	1282544	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696993	chr5	HAVANA	exon	1280273	1280453	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696994	chr5	HAVANA	exon	1279406	1279585	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696995	chr5	HAVANA	exon	1278756	1278911	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696996	chr5	HAVANA	exon	1268635	1268748	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696997	chr5	HAVANA	exon	1266579	1266650	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696998	chr5	HAVANA	exon	1260589	1260715	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696999	chr5	HAVANA	exon	1258713	1258774	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
697000	chr5	HAVANA	exon	1255402	1255526	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	

Supabase overview

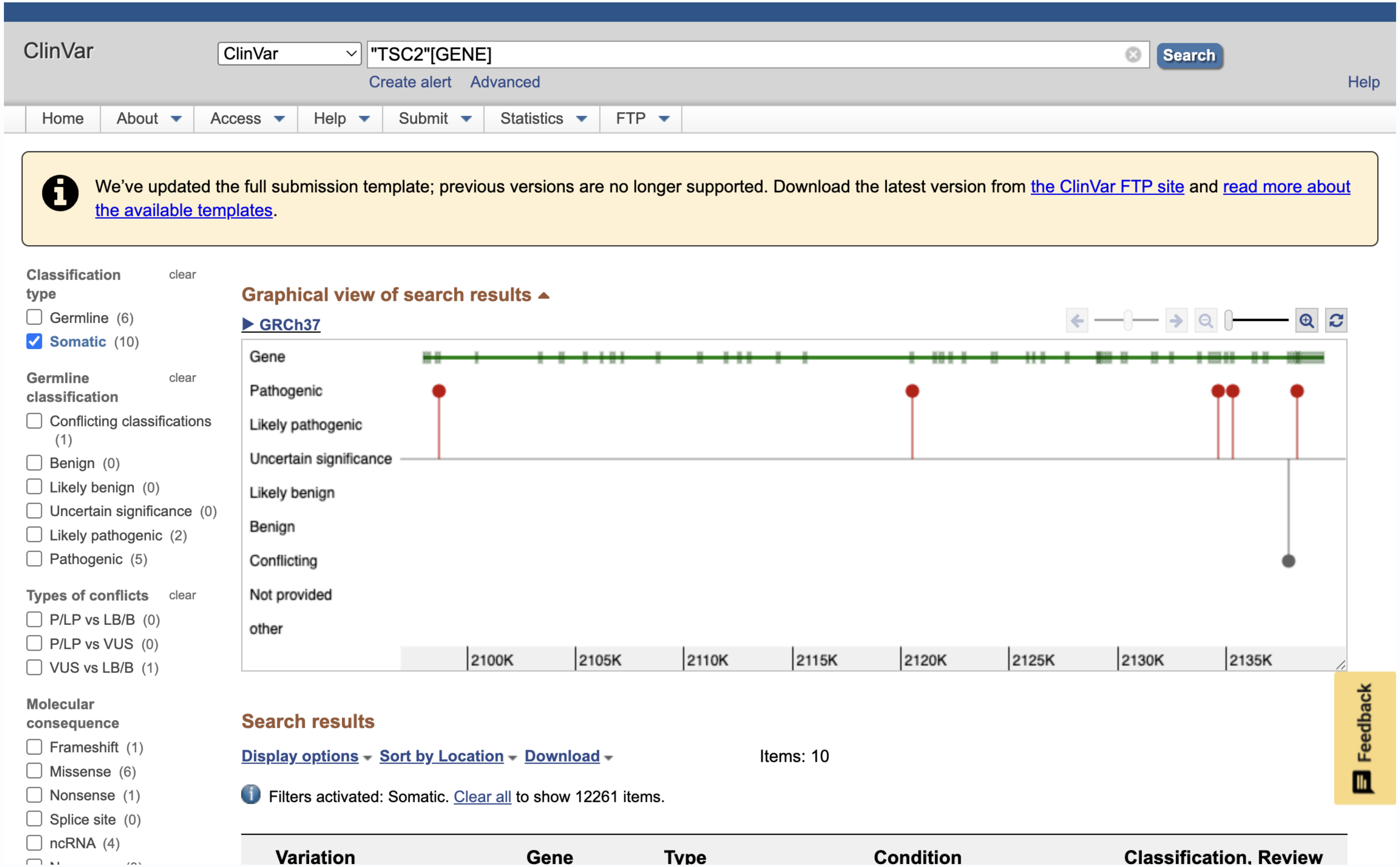


GeneBe



https://genebe.net/variant/hg38/chr17-43063873-G-A

Clinvar



https://www.ncbi.nlm.nih.gov/clinvar/

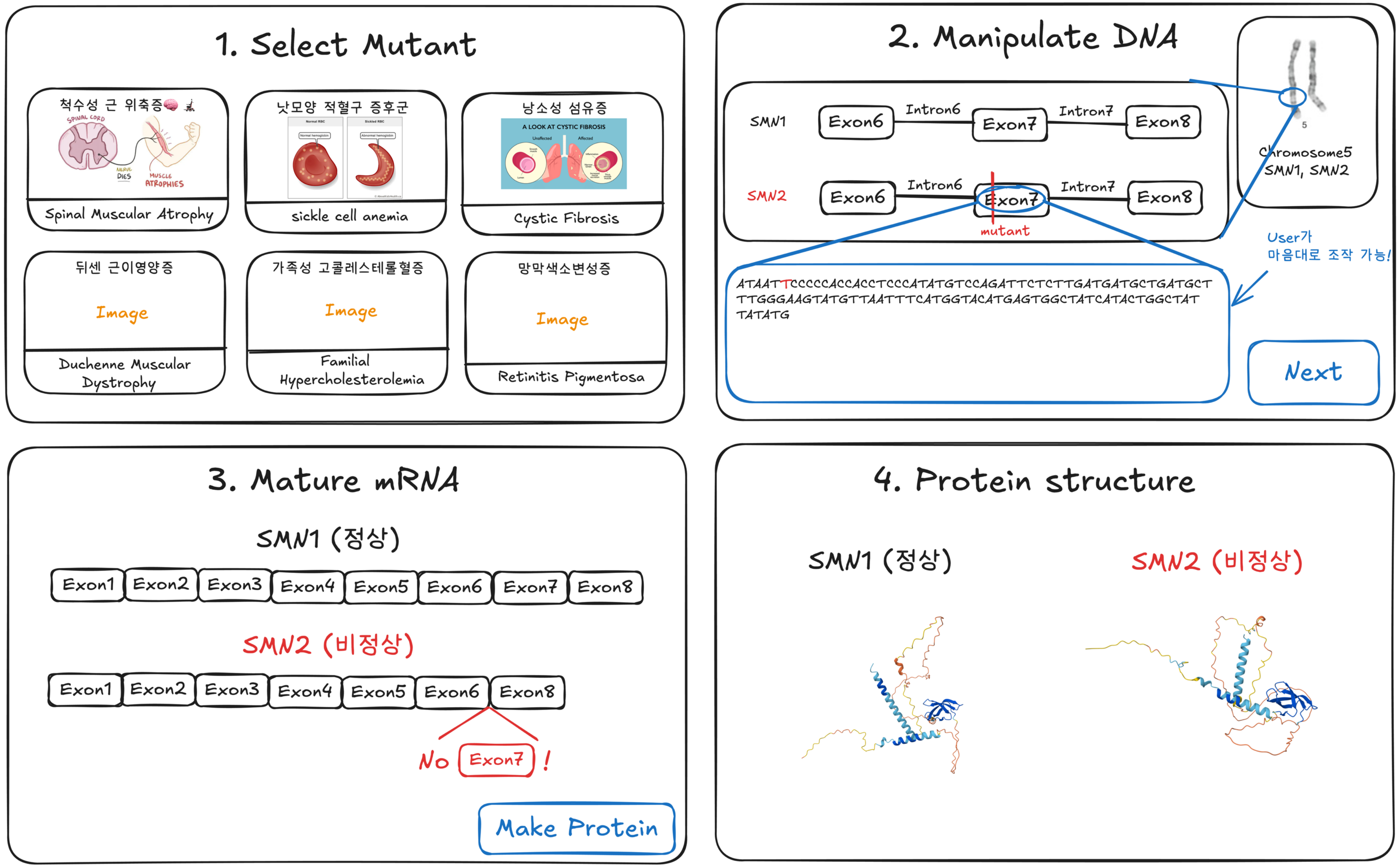
Clinvar

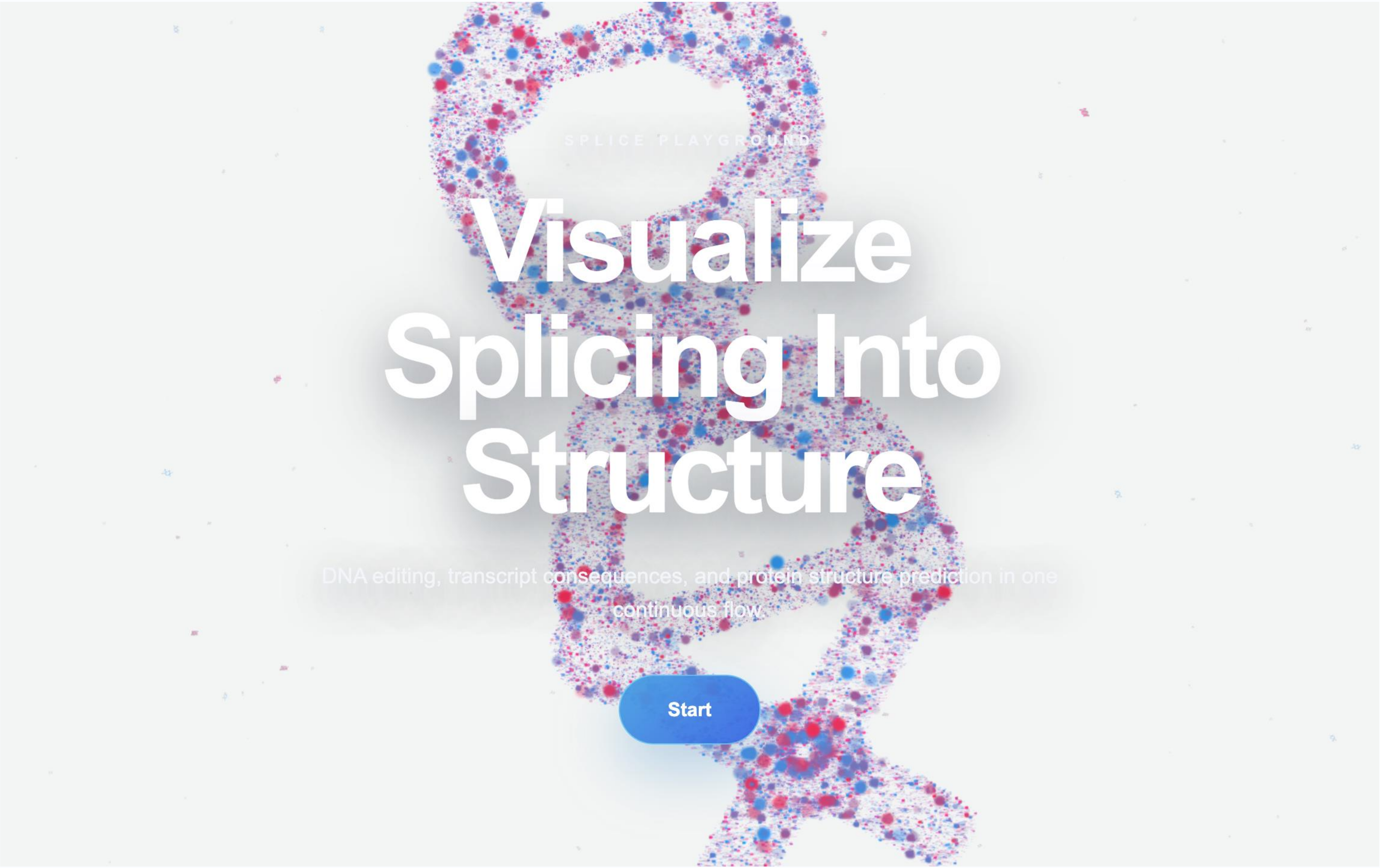
질병 선정에 도움을 주신 서울대 생명과학부 백대현 교수님,

박사과정 황현서 선배님, 포닥 전현성 선배님께 감사드립니다.

5. Progress

User Interface Review




SPLICE PLAYGROUND
Visualize Splicing Into Structure
DNA editing, transcript consequences, and protein structure prediction in one continuous flow.
Start

Progress

- Fixing minor errors and adding convenience features.

- The rest of the major framework is complete.

Reference

- Predicting Splicing from Primary Sequence with Deep Learning. *Cell*. 2019 Jan 24;176(3):535-548.e24. https://pubmed.ncbi.nlm.nih.gov/30661751/.
- Jumper, J., Evans, R., Pritzel, A. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021). *https://doi.org/10.1038/s41586-021-03819-2.*
- https://towardsdatascience.com/modeling-dna-sequences-with-pytorch-de28b0a05036/
- Some Lecture Notes from Prof. Daehyun Baek(Seoul National University, School of Biological Sciences)
- https://www.sciencedirect.com/science/article/pii/S187493991630275 9